



Network Protocol Attacks

Overview

This section covers hands-on exploitation of network protocols through man-in-the-middle (MitM) attacks, credential harvesting, and traffic analysis. Activities use industry-standard tools — Responder, Ettercap, and Wireshark — against a controlled lab environment.

Tools Used

- Responder — LLMNR/NBT-NS poisoning and SMB relay attacks
- Ettercap — ARP spoofing and MitM traffic interception
- Wireshark — Deep-packet capture and protocol analysis

Attack Simulation

SMB Relay Attack

An SMB relay attack was executed against the Metasploitable host at 192.168.1.4 using Responder. LLMNR/NBT-NS poisoning was leveraged to capture and relay NTLM authentication credentials. The attack successfully obtained an NTLM hash, recorded in the findings table below.

Attack ID	Technique	Target IP	Status	Outcome
015	SMB Relay	192.168.1.4	Success	NTLM Hash Captured

MitM Attack — ARP Spoofing with Ettercap

Ettercap was used to execute an ARP spoofing attack, positioning the attacker machine between the gateway and target host. This intercepted unencrypted traffic in real time, exposing credentials, session cookies, and DNS queries. Wireshark was used in parallel to capture and analyse the intercepted packet stream.



Summary

ARP spoofing via Ettercap successfully redirected LAN traffic through the attacker node, enabling passive capture of plaintext credentials and session data. DNS responses were also manipulated, demonstrating protocol-level trust exploitation in unsegmented networks.

Recommendations

- Disable LLMNR and NBT-NS on all workstations to prevent Responder-based poisoning.
- Enforce SMB signing on all hosts to block relay attacks.
- Deploy Dynamic ARP Inspection (DAI) on managed switches to prevent ARP spoofing.
- Use encrypted protocols (HTTPS, LDAPS, SMTPS) to protect credentials in transit.
- Segment the network using VLANs to limit lateral MitM exposure.